

Polyphenolics from acacia (Euterpe oleracea Mart.) and red muscadine grape (Vitis rotundifolia) protect human umbilical vascular Endothelial cell...

Endothelial anti-inflammatory effects of acacia (Ac) and red muscadine grape (Gp) polyphenolics have not been extensively investigated. It was hypothesized that polyphenolics from Ac and Gp exert comparable protective effects in human vascular endothelial cells (HUVEC) upon inflammatory stress. Furthermore, this study investigated whether microRNAs relevant to endothelial function might be regulated by Ac and Gp. Results showed that Ac and Gp (5-20 mg gallic acid equivalent/L) protected HUVEC against glucose-induced oxidative stress and inflammation. Glucose-induced expression of interleukin-6 and -8 was down-regulated by Ac and Gp at mRNA and protein levels. Upon lipopolysaccharide (LPS; 1 µg/L)-induced inflammation, Ac and Gp inhibited gene expression of adhesion molecules and NF-κB activation to similar extents, although Gp was more effective in decreasing PECAM-1 and ICAM-1 protein. Of the screened microRNAs, only microRNA-126 expression was found to be modulated by Ac and Gp as the underlying mechanism to inhibit gene and protein expression of VCAM-1.